

26) Title :-

Debugging an incorrect Rounding Algorithm in a DSP Program

Problem Statement :-

A DSP Program rounds every sample towards infinity instead of to the nearest integer. for samples :- $[-2.6, -1.4, 0.7, 1.8]$

Given Data :-

$[-2.6, -1.4, 0.7, 1.8]$

Implementation error :-

The Program uses rounding towards Positive infinity instead of rounding to the nearest integer

- Incorrect method

$$y = [x]$$

- correct method

$$y = \text{round}(x)$$

Incorrect outputs

I/P Samples

-2.6

-1.4

0.7

1.8

ceiling o/p

-2

-1

1

2

In correct output :- $[-2, -1, 1, 2]$

Correct rounded values

Sample 1

-2.6 $\rightarrow$ -3

Sample 2

-1.4 $\rightarrow$ -1

Sample 3

0.7 $\rightarrow$ 1

Sample 4

1.8 $\rightarrow$ 2

Corrected output is

$[-3, -1, 1, 2]$

Comparison Table :-

I/P	Incorrect o/p	Correct o/p
-2.6	-2	-3
-1.4	-1	-1
0.7	1	1
1.8	2	2

Root cause analysis

The implementation error occurs because the DSP Program applies the ceiling function, which always rounds numbers upward towards Positive infinity.

ex: -

$$\lceil -2.6 \rceil = -2$$

$$\text{round}(-2.6) = -3$$

Corrected Solution :-

Replace the ceiling operation :

$$y = \lfloor x \rfloor$$

with nearest integer rounding
operation

$$y = \text{round}(x)$$

$[-3, -1, 1, 2]$

Justification: -

*) Quantization converts continuous-valued samples into discrete integer values

*) Rounding to the nearest integer minimizes quantization error

*) Using the ceiling function introduces a positive bias, especially for negative values

Conclusion: -

Incorrect O/P: -

$[-2, -1, 1, 2]$

Corrected O/P: -

$[-3, -1, 1, 2]$

Incorrect samples: - -2.6